INCH-POUND

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30 September 1988
SUPERSEDING
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20 July 1972

MILITARY SPECIFICATION

MILLING MACHINE, VERTICAL, KNEE TYPE, SIZES 2 THROUGH 7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers vertical spindle, swivel head, plain table, knee and column, ram and turret type, electric motor driven, floor mounted milling machines with equipment and accessories specified herein.

1.2 Classification. The milling machines shall be of the following sizes, and the size to be furnished shall be as specified (see 3.4 and 6.2.1).

Size	Spindle drive (hp)	Minimum table width (inches)	Maximum table loading range (lbs)	Net weight with standard head, range (lbs)
2	2	9	650 to 850	2000 to 2800
3	3	10	750 to 1050	2600 to 3800
4	4	11	950 to 1350	3400 to 4600
5	4	11	1050 to 1550	3900 to 5100
6	4	11	1250 to 1850	4600 to 6200
7	5	13	1750 to 2550	5800 to 7500

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3417

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2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. The issue of these documents shall be the latest revision of the document on the date of the solicitation and may include title changes.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government documents and publications. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation and may include title changes.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be the latest revision of the document on the date of the solicitation and may include title changes.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators. (DoD Adopted)

ANSI/NFPA 70 - National Electric Code. (DoD Adopted)

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery. (1987 Edition) (DoD Adopted)

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues. (DoD Adopted)

ANSI B5.10 - Machine Tapers. (DoD Adopted)

ANSI B5.18 - Spindle Noses and Tool Shanks for Milling Machines. (DoD Adopted)

ANSI B11.8 - Drilling, Milling, and Boring Machines - Safety Requirements for the Construction, Care, and Use. (DoD Adopted)

ANSI/ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametrical Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 2000.A88 - Gear Classification and Inspection Handbook.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

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(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design. The milling machine shall be new (not a prototype) and one of the manufacturer's current models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, or repair. The machine shall consist of a base and column with a vertically movable knee on the front of the column and a slidable ram mounted in a rotatable turret on top of the column. A front-to-back movable saddle shall be mounted on the knee with a left-to-right slotted, movable table mounted on the saddle. An electric motor driven variable speed spindle assembly (head) with extendable quill shall be mounted vertically to one end of the ram, with adjustable swivel connections to the ram which allow the head to be tilted left-to-right or front-to-back. Lead screws with handwheels and measuring dials or scales shall provide precision movement of knee on column, saddle on knee, table on saddle, and quill extension. Multiple locking bolts shall hold securely the set positions of turret on column, ram in turret, and head on ram, while quick-release clamps shall provide for locking the table, saddle, and knee movements. Various means for attaching pieces to the table slots may be provided. Various control, power feed, digital readout and other options may also be provided. The complete machine is to provide drilling, tapping, positioning, and milling capabilities for metalworking.

3.1.1 Measurement systems. Unless otherwise specified (see 6.2.1), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the milling machine. In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376. However, the use of either system shall not prevent the use of standard commercially available components such as gears, bearings, seals, pins, and fasteners from either system of measurement, so long as they comply with the appropriate standards referenced herein and are readily available from domestic sources. All metric fasteners shall conform to FED-STD-H28.

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3.1.1.1 Measuring and indicating device graduations. Unless otherwise specified, (see 6.2.1). All measuring and indicating devices shall be graduated in either inch-pound units, or dual inch-pound and metric units.

3.1.2 Dials. All rotating dials which are used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.1.3 Digital readout system. Unless otherwise specified (see 6.2.1), a digital readout system shall be installed on the X and Y axes of travel. The system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to ensure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.0005" or smaller. The system accuracy shall be equal to the system resolution, verified upon installation and calibration of each system on the machine. The system repeatability shall be 0.0002 inch, (or half the resolution) measured over the full travel of each machine axis. Special requirements for the system components shall be as specified (see 6.2.1). The display shall indicate axis location in both the inch-pound and metric units. An input power source fluctuation of $\pm$ 10 percent shall not adversely affect the system operation. The system shall be provided with means of calibration, both electronic and manual, to the individual machine tool on which it is installed to compensation for the effects of the individual machine geometry errors.

3.1.3.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the digital readout system. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines in any multiple machine order.

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3.1.3.2 Electronic display console. The electronic display console for the digital readout system shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and shall be readable at a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's work station. The console shall be mounted to withstand shock and vibration generated by the machine. The connecting cable shall be armored and shall be installed so as not to interfere with the normal operations of the machine. The cable shall be securely fastened to the structure of the machine.

3.1.4 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize its intended use, performance, or design life. The reclaimed materials shall be reprocessed or recycled in a manner which will restore them to the same chemical composition and physical properties as virgin materials.

3.1.5 Energy efficiency. The milling machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.1.6 Controls. All mechanical, electrical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work stations.

3.1.7 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the milling machine, whether or not specifically required herein, shall comply with all requirements of 29 CFR 1910 and ANSI B11.8. If a conflict arises between 29 CFR 1910 and B11.8, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified in detail (see 6.2.1).

3.1.7.1 Safety guarding. Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial milling machine supplied to the commercial market shall be provided for the point of operation and all other parts of the milling machine that present safety hazards. The safety devices shall prevent unintentional contact with the guarded part and shall be removable to facilitate inspection, maintenance, and repair of the parts. The safety devices shall not interfere with the operation of the milling machine. When specific point-of-operation guarding that is dependent on workpiece size and configuration is required, the procuring activity shall specify the specific size and configuration of the guard required (see 6.2.1).

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3.1.8 Mercury restriction. The milling machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.1.9 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the milling machine.

3.1.10 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the milling machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation.

3.1.11 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each fluid lubricant reservoir shall have at least a 24-hour capacity and have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible. A central lubrication system shall be provided for all ways and axes screws on the knee and table. The lube reservoir and pump for this system shall be self-contained, readily accessible, and adjustable.

3.1.12 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts (except for hand-scraped and fitted parts such as ways and boxed bearings) shall be interchangeable without requiring modification. Replacement parts are those which are subject to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include, but not be limited to, motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Foreign-sourced replacement parts shall be interchangeable with domestically supplied components or be readily available from domestic inventories.

3.2 Construction. The milling machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the milling machine to its maximum rating and capacity without permanent distortion.

3.2.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as weldings, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.2.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

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3.2.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.2.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.2.5 Painting. Unless otherwise specified (see 6.2.1), the milling machine shall be painted in accordance with the manufacturer's commercial practice.

3.2.6 Threads. All threaded parts whether in US or SI units, used on the milling machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein. No other standards, regardless of claims, are acceptable.

3.2.7 Gears. All gears, shafts, and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 2000.A88 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears, shafts, and pinions shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears, shafts, and pinions in the drive train shall be hardened and ground steel. Surface hardness shall be not less than Rockwell C-48.

3.2.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.3 Components. The milling machines shall consist of a ram, ram turret, table, saddle, knee, column and base, swivel head, spindle, and such other features as are necessary to meet the performance requirements specified herein.

3.3.1 Ram. The ram shall be designed for mounting the plain or universal swivel head, as applicable. The ram shall have matching slides machined and fitted to the turret ways. Means shall be provided for clamping and locking in any desired position within the rated ram travel. Ram positioning shall be by manual mechanical means.

3.3.2 Ram turret. The ram turret shall be machined and fitted to the top of the column. The turret shall have means for clamping in any position within a 180-degree horizontal plain. The turret shall be provided with suitable markings to indicate zero swivel. The top of the turret shall be designed with dovetail or square lock ways with matching slides on the ram.

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3.3.3 Table. The table shall be of one-piece construction. The bottom of the table shall be machined and fitted to ways of the saddle. It shall have a locking arrangement for clamping to the saddle. The table shall be of the trough type with coolant channels and a drain hole fitted with a return line to the reservoir. Table surface shall be machine finished and provided with not less than three equally spaced T-slots. The front side of the table shall be a finished surface with a T-slot running the full length of the table for mounting adjustable stops. All T-slots shall be in accordance with ANSI B5.1. Two adjustable table stops shall be furnished. Manual longitudinal table feed shall be by means of a feed screw operated by a handwheel or crank and shall be fitted with a dial graduated in thousandths of an inch with every fifth graduation extended and every tenth graduation numbered until return to zero. When specified (see 6.2.1), power feed to the table shall be furnished. Means shall be provided to prevent overtravel of the table feed mechanism.

3.3.4 Saddle. The saddle shall have strength and proportions suitable for supporting the table and its maximum rated load capacity. The saddle shall be equipped with tapered adjustable gibs and guideway wipers or other suitable means for preventing dust and chips from entering ways. Bearing surfaces and ways shall be machined and scraped to fit mating parts. The saddle shall have means for lubrication of all knee, saddle, and table bearing surfaces and moving mechanisms in the saddle. Manual saddle cross feed shall be provided. The feed screw shall be heat-treated steel with acme or square threads. When specified (see 6.2.1), power feed to the saddle shall be furnished. The cross feed handwheel shall be automatically disengaged when power is engaged.

3.3.5 Knee. The knee shall be machined and fitted to the column ways with adjustable gibs. The knee shall support the saddle and table and shall have vertical movement by means of an elevating screw operated by a crank or handwheel mechanism. When specified (see 6.2.1), power elevation of the knee shall be supplied. (See table I and note that power knee elevation is not available on all models.)

3.3.6 Column and base. The column and base construction shall minimize vibration and assure the accuracies and performance specified herein. The front of the column shall have machined and fitted ways for the knee. The top of the column shall be machined and fitted to the ram turret. The periphery of the column top shall be marked to duplicate the zero setting marks on the ram turret. The base shall have provisions for securing the machine to the floor or foundation.

3.3.7 Swivel head. Each machine shall have a swivel head with one or more graduated scales or dials to indicate the angular position of the head in the plane, or planes, in which it moves. Unless otherwise specified (see 6.2.1), the head shall be provided with a mechanism for fast manual movement of quill, a handwheel or knob for slow manual quill movement, a micrometer depth stop, a quill locking device, and power quill feed.

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3.3.8 Spindle. The spindle shall be of heat-treated alloy steel journalled on precision type ball or roller bearings. Means shall be provided for axial adjustment of the bearings. The spindle nose shall be precision ground. It shall be furnished with the standard taper (see table II) and conform to ANSI B5.10 and B5.18 unless otherwise specified (see 6.2.1). The Number 40 or larger taper shall not be furnished in spindles with quill diameters less than 4.25 inches.

3.3.9 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant to the cutting tool over the entire work area.

3.3.10 Electrical system. The electrical system shall conform to ANSI/NFPA 79. Unless otherwise specified (see 6.2.1), each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The unit shall be initially wired for operation on 460 volts.

3.3.10.1 Motors. Motors shall be rated for 60 minute minimum duty and shall have ball or roller bearings of the sealed, balanced, and permanently lubricated type. Alternating-current (AC) motors shall be designed to operated on 60-Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall meet the requirements of ANSI/NEMA MG-1 and ANSI/NFPA-79.

3.3.10.2 Control circuit voltage. Auxiliary control circuits shall be isolated from the input power supply by a low-voltage transformer having a secondary voltage of 120 volts or less.

3.3.10.3 Hour meter. When specified (see 6.2.1), the milling machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the milling machine.

3.3.10.4 Electrical load indicator. When specified (see 6.2.1), the machine shall have a meter to indicate the electrical load on the spindle drive motor. The dial shall indicate either amperes or percent of rated full-load current, and shall be clearly marked to indicate the rated reading for continuous operation of the machine. The meter shall be accurate within 2 percent of full-rated load. The meter shall be clearly readable from the operator's normal position.

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3.4 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the milling machine shall be not less than specified in table I for the machine size specified, and table II for the head size specified.

3.5 Performance. The milling machine shall meet the performance test requirements when tested in accordance with 4.7.3 through 4.7.4.

3.5.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.6 Alignment tolerances. Each milling machine shall meet the alignment tolerances specified in table III.

3.7 Worklight. Unless otherwise specified (see 6.2.1), a worklight shall not be provided.

3.8 Standard equipment. All standard equipment required to properly operate the milling machine (such as special tools) shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2.1). All repair parts shall be assigned a part number identified in a repair parts manual.

3.11 Marking on instruments, control panels and plates. All words on instruments, control panels, and plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion-resistant. Unless otherwise specified, all symbols, units, and special characters, shall be in the inch-pound system of measurement.

3.11.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity
Military or Federal specification number for each lubricant
Size and type of each filter

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation

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shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total full-load amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture
Ampere rating of largest motor or load
Short-circuit interrupting capacity of the machine overcurrent protective device where furnished

3.12 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language and in the inch-pound system of measurements.

3.13 Workmanship. Workmanship of the milling machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (see 6.5).

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.4).

b. Acceptance test (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in an indoor facility with ambient conditions of temperature: 40°F to 105°F; humidity: 20 percent to 90 percent relative humidity; and atmospheric pressure: 15.0 to 13.0 psi.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied by the contractor to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspections shall consist of the examinations in 4.6, the tests in 4.7, and the inspection in 4.8. Failure of the item to pass the examination, the tests, or the inspection shall be cause for rejection.

4.5 Acceptance test. Acceptance test shall be performed on the milling machine to ensure conformance with this specification. Unless otherwise specified (see 6.2.1), the acceptance test shall consist of examinations in 4.6.1 through 4.6.5 and all tests in 4.7. Failure of the milling machine to pass one or more tests or examinations will be cause for rejection. Unless otherwise specified (see 6.2.1), the acceptance test shall be performed at the user facility. This requirement shall be waived only when authorized by the user facility in writing.

4.6 Examination. The milling machine shall be examined to determine compliance with all requirements of this specification including compliance with all referenced standards.

4.6.1 The main machine nameplate shall be examined for compliance with 3.11.2.

4.6.2 The lubrication nameplate shall be examined for compliance with 3.11.1.

4.6.3 Each electric motor shall be examined for a permanently mounted nameplate having all data required by Section 1-4 and 16-5 of NFPA 79, 1987 Edition.

4.6.4 The machine wiring shall be examined for compliance with sections 1-4, 13-1 through 13-6 and 14-1 through 14-4 of NFPA 79, 1987 Edition. The presence of any conductors other than soft-annealed copper, the lack of color coding, or the lack of marked wiring and terminals shall be cause for rejection. This requirement shall not be waived without authorization from the user facility in writing.

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4.6.5 The wiring diagrams, schematics, panels, and instruments shall be examined for compliance with Sections 1-4, 2-1 through 2-8, 11-1, and 11-2 of NFPA 79, 1987 Edition. Use of symbols other than those approved, items not marked, items marked differently than the schematic or wiring diagram, or lack of required schematics and diagrams shall be cause for rejection.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Alignment tolerance test. Each machine shall be tested to determine that it complies with the alignment tolerance requirements of 3.6, after testing of 4.7.3 through 4.7.4, and without special adjustments on tightening of gibs or guides. Any adjustments or tightening required shall be cause for rejection.

4.7.3 Performance tests. At the option of the procuring agency, each machine shall be subjected to tests in accordance with specific requirements designated by the activity in lieu of or in addition to the following tests (see 6.2.1). All performance tests shall be completed without making any adjustment to or performing any maintenance on the machine.

4.7.3.1 Horsepower, milling accuracy, and tool chatter tests. Each machine shall be tested by machining a workpiece of SAE 4140 steel. The workpiece shall be not less than 16 inches long, 6 inches wide, and 3 inches thick. The workpiece may be drilled to facilitate clamping. One of the two surfaces having the largest area shall be ground flat and mounted in contact with the table. Roughing cuts shall be made to mill metal off the horizontal surface and off two adjacent vertical surfaces of the workpiece. The roughing cuts shall be made at sufficient depth and feed rate to load the spindle drive motor to not less than maximum duty rated electrical input. The type of cutting tool, depth of cut, spindle speed, and feed rate shall be determined by the manufacturer, provided the required loading is achieved. The resultant metal removal rate shall not be less than 0.65 cubic inches per minute per the rated horsepower of the spindle drive motor. The surface of the workpiece shall show no evidence of chatter and there shall be no evidence of chatter, stalling, or permanent distortion during or after the test. Finishing cuts shall be made on the horizontal and vertical surfaces. The finished surfaces shall meet the following accuracies:

4.7.3.2 Squareness. The two milled sides shall be tested for squareness. The sides shall be square with each other within 0.001 inch per foot.

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4.7.3.3 Flatness. The two milled sides shall be tested for flatness. The sides shall be flat within 0.001 inch per foot.

4.7.3.4 Parallelism. The finished horizontal surfaces shall be tested for parallelism. The top surface shall be parallel to the bottom surface within 0.001 inch per foot.

4.7.4 Climbing milling test. Each machine shall be tested in accordance with the manufacturer's standard climb milling test, provided that the test proves the capability of the machine to meet all of the requirements for which it was designed.

4.7.5 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.2.8. Testing procedures shall be as given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packaging, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The milling machines covered by this specification are to be used for conventional milling operations on a wide range of parts, primarily metal.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. If the milling machine is required to be configured in a specific measurement system (inch-pound or metric), state required system (see 3.1.1).

d. If measuring and indicating devices are required to be graduated in a specific measurement system (inch-pound or metric), state required system (see 3.1.1.1)

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e. Additional safety and health requirements, if required (see 3.1.7).

f. Digital readout system, if different; specify axes of travel and any special requirements (see 3.1.3 and 3.1.3.1).

g. Painting, if different (see 3.2.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.2.8).

i. Power table feed, if required (see 3.3.3).

j. Power saddle feed, if required (see 3.3.4).

k. Power knee elevation, if required (see 3.3.5).

l. Swivel head, if different (see 3.3.7).

m. Spindle taper, if different (see 3.3.8).

n. Electrical system, if different (see 3.3.10).

o. Motors, if different (see 3.3.10.1).

p. Hour meter, if required, (see 3.3.10.3).

q. Electrical load meter, if required (see 3.3.10.4).

r. Size and capacity, if different (see 3.4).

s. Additional performance features, if required (see 3.5.1).

t. Worklight, if required (see 3.7).

u. Specify and fully describe optional equipment, if required (see 3.9).

v. Specify applicable repair parts, if required (see 3.10).

w. Lubrication plate, if different (see 3.11.1).

x. Nameplate, if different (see 3.11.2).

y. Inspection conditions, if different (see 4.3).

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z. Quality conformance inspection, if different (see 4.4).

aa. Acceptance test, if different (see 4.5).

bb. Location of acceptance test, if different (see 4.5).

cc. Performance tests, if different (see 4.7.3).

dd. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

ee. Specific point-of-operation guarding, if required (see 3.1.7.).

6.2.2 Contract data requirements. Required technical data (operating manuals, part lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 Cross reference of classification changes. The milling machine classifications covered by this specification have been changed: from sizes 1 through 5 (based on spindle horsepower) with classes 1 and 2 (based on table width); to machine sizes 2 through 7 (based on sequential progressions of spindle horsepower, table width, table loading and net weight with standard head) (see 1.2 and table I). Head sizes 2 through 5 (based on standard spindle horsepower) are classified separately (see table III).

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Inspection facilities. The contracting officer should be notified by each contractor of the location where inspection tests are to be performed. It is recommended all inspections be performed within the limits of the continental United States.

6.7 Safety health requirements. Standard 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The using activity is advised to analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.8 Training Training required by the using activity shall be provided as specified in the contract.

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6.9 Warranty Warranty requirements shall be as specified by the using in the solicitation. It is recommended that the warranty be required to begin after installation and final acceptance tests and examinations have been successfully completed.

6.10 Concerns to note. A number of the machines offered under this specification maybe manufactured in foreign countries and may not meet required U.S. Standards. Requirements for Standards, espcially NFPA 79 and screw thread standards should not be waived or ignored.

The quality of foreign sourced machines may vary more widely, spare parts may be more expensive and have longer lead times, tech data maybe in a foreign language and panel symbols hamper the efficiency of the machine's operator. Due to one or a combination of these factors, the operating and maintenance costs may increase, sometimes dramatically.

6.11 Part of Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:

M	XXXXX	-	X	X
Prefix to indicate military specification	Specification number-No revision letter	An Alph Code may replace the dash	Size from Table I	Size from Table II

6.12 Subject term (keyword) listing.

Knee type
Machine Tools
Milling Machine
Ram Type
Turret type
Vertical

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TABLE I. Sizes and capacities.

Characteristics	Size 2	Size 3	Size 4
Table working surface			
Length x width, inches			
Standard table size	42 x 9	48 x 10	58 x 11
Optional table size #1	48 x 9	44 x 10	52 x 12
Optional table size #2	36 x 9	49 x 12	----
*Head size, standard var, speed	2 HP	3 HP	4 HP
Optional step pulley heads	1/2, 1-1/2, 2, 3 HP	2 HP	2 HP
Optional variable speed heads	1, 1-1/2, 2, 3 HP	----	3 HP
Longitudinal table travel, inches			
With standard table (with power feed)	30 (26-1/2)	31 (29)	36 (30)
With optional table #1 (with power feed)	36 (32-1/2)	31 (31)	31 (31)
With optional table #2 (with power feed)	24 (20-1/2)	31 (31)	----
Cross travel, inches (with power feed)	12	15	15
Vertical travel, inches (with power feed)	16	16	16
Table power feeds (number of) IPM range			
Longitudinal standard	----	(12) 1 to 20	(12) 1 to 20
Longitudinal optional #1	(inf) .3 to 10	----	----
Longitudinal optional #2	(inf) 1 to 35	----	----
Cross standard	----	(12) 1 to 20	(12) 1 to 20
Cross optional #1	(inf) .3 to 10	----	----
Cross optional #2	(inf) 1 to 35	----	----
Vertical standard	----	----	----
Vertical optional	(inf) 1 to 35	(1) 42-5/16	(12) 1 to 10
Throat dimensions			
Spindle axis to body, inches	7 to 18	7 to 30	8 to 24
Nose of quill to table, inches	3 to 18	3 to 17	1 to 15
Height, inches	83 max.	91 max.	72 max.
Recc. max. table load, lbs.	650 to 850	750 to 1050	950 to 1350
**Net wt. with standard head, and table, lbs	2000 to 2800	2600 to 3800	3400 to 4600

NOTE: When a performance range is shown, the required performance range is from the stated minimum or less to the stated maximum or greater.

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TABLE I. Sizes and capacities. continued.

Characteristics	Size 5	Size 6	Size 7
Table working surface			
Length x width, inches			
Standard table size	58 x 11	58 x 11	63 x 13
Optional table size #1	----	----	----
Optional table size #2	----	----	----
*Head size, standard var. speed	4 HP	4 HP	4 HP
Optional step pulley heads	----	----	5 HP
Optional variable speed heads	1, 1-1/2, 2 HP	1, 1-1/2, 2 HP	
Longitudinal table travel, inches			
With standard table (with power feed)	33 (32)	--- (30)	46 (45)
With optional table #1 with power feed)	----	----	----
With optional table #2 with power feed)	----	----	----
Cross travel, inches (with power feed)	15	15	(12)
Vertical travel, inches (with power feed)	12	16 (16)	(16)
Table power feeds (number of) IPM range			
Longitudinal standard	(inf) 1 to 35	(inf) 1 to 24	(18) 1 to 20
Longitudinal optional #1	(inf) .3 to 10	----	----
Longitudinal optional #2	----	----	----
Cross standard	----	(inf) 1 to 24	(18) 1 to 20
Cross optional #1	----	----	----
Cross optional #2	----	----	----
Vertical standard	----	(inf) 1 to 12)	(18) 1 to 20
Vertical optional	----	----	----
Throat dimensions			
Spindle axis to body, inches	6 to 23	11 to 26	7 to 37
Nose of quill to table, inches	4 to 16	1 to 17	4 to 20
Height, inches	83 max.	93 max.	72 max.
Recc. max. table load, lbs.	1050 to 1550	1250 to 1850	1750 to 2550
Net wt. with standard head, lbs	3900 to 5100	4600 to 6200	5800 to 7500

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Table II. Standard mill heads - sizes and capacities.

Characteristics	Size 2	Size 3	Size 4	Size 5
Motor HP.	2	3	4	5
Spindle speeds (NOTE 1) Var. speed range, RPM	60 to 4000	60 to 4000	60 to 3500	20 to 2200
Quill dia., min., inches	3.38	3.38	4.25	4.25
Quill feeds				
Number of Range, inches per revolution	3 .0015 to .006	3 .0015 to .004	3 .0015 to .004	3 .0015 to .006
ANS taper, standard (NOTE 2)	#30, R-8	#30, R-8	#40	#40
Head swivel, L. to R., degrees	90 -0- 90	90 -0- 90	90 -0- 90	90 -0- 90
Head swivel, fr. to b., degrees	45 -0- 45	45 -0- 45	30 -0- 30	None
Collet capacity, when furn., inches	.75	.88	1.00	1.00

NOTE 1: Infinitely variable speed drives are approved as substitutes for a finite no. of speeds, provided the speed range is duplicated within 10 percent.

NOTE 2: R-8 taper is included due to wide commercial usage, though not covered by the referenced ANSI standards.

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User activity:
Navy - MC

Preparing activity:
DLA - IP

Project (3417-0173)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-80016C	Milling Machine, Vertical, Knee Type, Sizes 2 through 7

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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